

Curriculum Vitae/Resume

Gabriel Kosmacher

<https://kennykos.github.io/>

Austin, Texas, United States

+1 773-986-0852

Education

University of Texas at Austin, Austin, TX
Degree: Computational Science, Engineering, and Mathematics PhD
Concentration: Computational and Applied Mathematics (CAM)
Advisor: George Biros

August 2023-Present

University of Illinois at Urbana-Champaign, Champaign, IL
Degree: Mathematics and Computer Science BSLAS; Highest Distinction
Minors: Computational Science and Engineering

August 2019-May 2023

Publications

Kosmacher, Gabriel, Dillon Max, et al. (Jan. 2026). “The Role of Host Immunity and the Environment in Seasonal Disease Dynamics”. In: *Bulletin of Mathematical Biology* 88.2, p. 18. ISSN: 1522-9602. DOI: [10.1007/s11538-025-01582-3](https://doi.org/10.1007/s11538-025-01582-3).
Kosmacher, Gabriel and K. Shankari (July 2024). “Evaluating the Interplay between Trajectory Segmentation and Mode Inference Error”. en. In: *Transportation Research Record: Journal of the Transportation Research Board* 2678.7, pp. 32–49. ISSN: 0361-1981, 2169-4052. DOI: [10.1177/03611981231208154](https://doi.org/10.1177/03611981231208154).
Du, Yu et al. (Oct. 2023). “Packing Densities of Delzant and Semitoric Polygons”. en. In: *SIGMA. Symmetry, Integrability and Geometry: Methods and Applications* 19, p. 081. ISSN: 18150659. DOI: [10.3842/SIGMA.2023.081](https://doi.org/10.3842/SIGMA.2023.081).

Preprints

Kosmacher, Gabriel, Ziyu Du, et al. (June 2026). *A performance portable fast Ewald summation for Stokes flow*. preprint. arXiv: [2606.19059](https://arxiv.org/abs/2606.19059) [math.NA]. URL: <https://arxiv.org/abs/2606.19059>.
Kosmacher, Gabriel, Eric T. Phipps, and Sivasankaran Rajamanickam (Oct. 2025). *A Performance Portable Matrix Free Dense MTTKRP in GenTen*. preprint. arXiv: [2510.14891](https://arxiv.org/abs/2510.14891) [cs.MS]. URL: <https://arxiv.org/abs/2510.14891>.

Presentations

Kosmacher, Gabriel (May 2025). “Performance Portable Fast Ewald Summation”. In: *High Performance Software Foundation Conference 2025*. Chicago, Illinois.

Research Experience

University of Texas at Austin August 2023-Present
Dr. George Biros
Working on a performance portable Spectral Ewald implementation for particle-in-cell Stokes flow.

- **Developing a novel particle-to-grid algorithm** to be used in the *particle2grid* stage of the Spectral Ewald method that utilizes **data locality** and **parallelism** to achieve **near-peak performance** on modern machines.
 - Algorithm development in **PyKokkos**, a **performance portable** python framework. The algorithm will be released as part of a planed **open source** python package for the Spectral Ewald method.
 - Performance profiling with the **NSIGHT Compute** suite of tools for runs using NVIDIA’s A100 GPUs on the TACC Lonestar 6 supercomputer. These results are compared with **cache aware** pen-and-paper performance modeling.
 - Invited to AMD tools workshop at the University of Oregon port the algorithm onto **HIP** architecture and tune for performance on AMD Instinct™ series MI300A devices.

Sandia National Laboratory
Dr. Eric. T. Phipps, Dr. Siva Rajamanickam

August 2025-October 2025

- Developed a **performance portable matrix-free MTTKRP** kernel for **dense tensor decompositions**. The implementation is the default for Sandia’s GenTen package <https://github.com/sandialabs/GenTen.git>. A preprint of the work is available on arxiv <https://arxiv.org/abs/2510.14891>.
- Developed linear algebra tutorials for the Cerebras wafer-scale engine in the tungsten & paint programming languages.

National Renewable Energy Laboratory

June 2022-August 2022

Dr. K. Shankari

Worked on *Evaluating The Interplay Between Trajectory Segmentation and Mode Inference Errors*

- **Introduced a framework** to evaluate accuracy of trip length computations and mode inference for continuous mode-segmented trajectories on groups of trips.
 - Developed a temporal alignment algorithm to classify temporal and spatial errors in a single metric and implemented the algorithm in python <https://github.com/kennykos/mobilitynet-analysis-scripts>
 - Applied the framework to the NREL OpenPATH pipeline using MobilityNet, a public dataset containing information from three *artificial timelines* that cover 15 different travel modes.
 - Evaluated travel data collected on smartphones on android and iOS operating systems that was post-processed by different machine learning algorithms.
- **Co-authored a manuscript** currently Published in the Transportation Research Record: <https://doi.org/10.1177/03611981231208154>.
- **Results presented** at the Transportation Research Board Annual Meeting 2023.

University of Illinois Department of Mathematics

Jan 2022-May 2023

Dr. Zoi Rapti

Worked on *Seasonality and Immunity in Disease Dynamics*

- **Co-designed and Co-developed a dynamical model** with another undergraduate student and Professor Rapti to investigate *Daphnia dentifera* disease dynamics.
- **Presented results** at the University of Illinois Chicago Undergraduate Mathematics Symposium 2022 and the University of Illinois Urbana-Campaign Undergraduate Research Symposium 2022.

Illinois Geometry Lab

January 2021-May 2022

Dr. Joey Palmer

Worked on *Toric and Semitoric Packing Capacities*

- **Investigated packing capacities** with a team of undergraduate student and Professor Palmer to exactly compute packing capacities.
 - Developed an algorithm to explicitly compute toric packing capacities and implemented the algorithm in python https://github.com/kennykos/Semi-toric_Packing_Capacity.
 - Solved the equivariant semitoric perfect packing problem.
- **Co-authored a manuscript** currently Published in Symmetry, Integrability and Geometry: Methods and Applications: <https://doi.org/10.3842/SIGMA.2023.081>.
- **Presented results** at University of Illinois Urbana-Campaign Illinois Geometry Lab Poster Presentation Spring 2021, Fall 2022, Spring 2022.
- **Received** 2022 Illinois Geometry Lab Outstanding Research Award.

Grants/Awards

CSEM Fellowship

2024-2028

Illinois Geometry Lab Outstanding Research Award

2022

Americorps Education Award

2021

Heery Scholarship Recipient

2020-21

Work

Experience

University of Illinois Department of Computer Science

October 2022-May 2023

CS 450 Numerical Analysis Course Assistant

- **Graded** Mathematical & CS theory homework problems for an advanced undergraduate/graduate course.

National Renewable Energy Laboratory

June 2022-August 2022

Science Undergraduate Laboratory Internship

- Developed continuous mode-segmented trajectory framework (see research section above).
- Worked with a **team of 3 interns** on statistical methods for trajectory **error propagation**.
- Participated in numerous Department of Energy **professional development activities and workshops**.

University of Illinois Department of Mathematics

August 2021-May 2022

Mathematics and Statistics Student Support Center

- Hosted **drop in office hours** for all mathematics courses up to Calculus II.

CSEM Student Forum

August 2024-June 2025

Co-host

- General organization (including speaker invitation) for a weekly seminar series given by current CSEM graduate students to their peers. The aim of the forum is to expose students to each other's research, encourage collaboration, and provide opportunities to practice presentation skills.

UT Austin Green Fund

January 2024-June 2025

Committee Member

- Member of a small student-faculty committee that reviews proposals and awards for the UT Austin Green Fund, a competitive grant program funded by a tuition allocation to support sustainability related projects and initiatives proposed by university students, faculty or staff.

Chicago Pre-College Science and Engineering Program

October 2022-December 2022

STEM Mentor

- Assisted in the teaching of a data-science curriculum at Kenwood Academy high school in Chicago for 11th and 12th graders.
- Developed a plant biology and environmental sustainability curriculum for 2nd graders which will be taught at the Urbana Neighborhood Connections Center.